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SCHOOL OF COMPUTING
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Real-Time Cyber-Physical Threat Detection for Industrial Control Systems

23CSE399 – Project Phase 1 (Panel Review 1)
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1 Problem Definition & Originality

1.1 Problem Statement

Industrial Control Systems (ICS) in critical sectors like Oil & Gas and Water Treatment operate under strict real-time and safety constraints yet face increasingly sophisticated cyber threats. Traditional security solutions often introduce latency, require significant resources, or fail to detect stealthy and zero-day attacks. This challenge is intensified by legacy PLCs and SCADA devices with limited CPU and memory. As breaches can cause serious safety, operational, and financial consequences, there is a need for a lightweight, real-time anomaly detection system capable of operating within minimal resource limits, identifying unknown threats, and incorporating physics-aware analysis.

1.2 The Challenge

Industrial Control Systems (ICS) face a unique set of constraints that render traditional IT security solutions ineffective:

- **Strict Real-Time Constraints:** Operations occur in milliseconds; any security delay can trigger safety shutdowns.
- **Legacy Infrastructure:** Most PLCs run on decades-old hardware with insufficient CPU/RAM to host modern security agents.
- **Stealthy & Complex Attacks:** Advanced Persistent Threats (APTs) often use valid protocols to hide malicious intent (e.g., Stuxnet), evading standard firewalls.

1.3 Proposed Approach

To address these specific challenges, we propose an **Intrusion Detection System** with the following characteristics:

1. **Real-time detection**
2. **Minimal resources**
3. **Detects zero-day threats**
4. **Physics-Aware**

2 Literature Survey & Gap Analysis

Paper / Focus	Advantages	Limitations
Karacayılmaz et al. *AI-based IIoT* [1] (2024)	Adaptive to new threats; high accuracy using advanced AI models.	"Black-box" nature lacks explainability; high computational cost for legacy PLCs.
Murugesan et al. *Statistical Approach* [2] (2024)	Handles missing data effectively; mitigates early attack discrimination.	Designed for Smart Grids, not Oil & Gas; complex statistical tuning required.
Mohammed et al. *Field flooding* [3] (2023)	Specifically tailored for O&G infrastructure; effective against DoS.	Narrow scope (flooding only); may miss subtle Logic Injection attacks.
Barsha & Hubballi *State Transition* [4] (2024)	Detects semantic attacks by validating the *sequence* of operations.	Manual state modeling is complex; struggles with high-speed, concurrent states.
Athar et al. *Dual-Layer Attestation* [5] (2025)	Strong integrity verification; prevents firmware tampering.	High overhead for real-time loops; requires specific architecture changes.
Srivastava et al. *DDoS in SCADA* [6] (2024)	Critical protection against Availability attacks (DDoS) ensuring uptime.	Focuses on Network Layer (volumetric); misses Application Layer (logic) attacks.

Research Gap: Existing solutions exhibit at least one of the following limitations:

1. Computational latency affecting real-time constraints.
2. Narrow focus on specific attack types.
3. Implementation challenges for legacy systems.
4. Inadequate handling of advanced stealth threats like Digital Twin desynchronization, which evades AI-based monitoring.

3 Uses of ICS in Critical Infrastructure

- Oil & Gas Industry
  - Drilling and well control (pressure, temperature, vibration)
  - Blowout Preventer (BOP) safety systems
  - Pipeline monitoring and leak detection via SCADA
  - Compressor and pump station automation
- Water Treatment Plants
  - Monitoring water levels, flow rates, and pressure
  - Automated chemical dosing (chlorine, pH control)
  - Pump and valve control for distribution systems
  - Detection of contamination or sensor anomalies
- Power Grid / Energy Sector
  - Substation automation and protection relays
  - Load balancing and frequency regulation
  - Turbine, generator, and transformer monitoring
  - Fault detection and grid stability control

4 SCADA Vulnerabilities & Threats

1. **Man-in-the-Middle (MitM) [1]**
   SCADA systems, especially in substations, often communicate via unencrypted protocols.
   Threat: Attackers intercept and modify control signals in transit.
   Impact: Valid commands are replaced with malicious ones without the operator's knowledge.

2. **Architectural Flaws [3]**
   Many legacy architectures lack internal segmentation (flat networks).
   Threat: Once an attacker breaches a single node (e.g., HMI), they have unrestricted access to all connected PLCs.
   Impact: Rapid propagation of malware across the entire facility.

3. **State Manipulation [1, 5]**
   Attackers can decouple the HMI display from physical reality.
   Threat: Reporting a "Safe/Closed" state to the operator while the valve is actually "Open."
   Impact: Operator blindness leading to physical damage (e.g., Stuxnet centrifuge attack).

4. **Digital Twin Desynchronization**
   Modern ICS uses Digital Twins for predictive monitoring.
   Threat: Attackers feed false data to the Twin to ensure it matches the "expected" behavior.
   Impact: Hides the malicious activity from AI/ML detection systems relying on the Twin.

5 Historical Cyber Attacks

Year	Attack Type	Target System	How They Attacked	Key Defenses
2021	Remote PLC manipulation	Florida water treatment plant (Oldsmar)	Attacker accessed plant via exposed remote access software (TeamViewer).	Disable unnecessary remote access, enforce MFA, network segmentation, continuous monitoring of PLC logic changes. [7]
2022	Industrial malware (ICS wiper)	Ukrainian energy infrastructure	Industroyer2 malware deployed against high-voltage electrical substations, issuing malicious commands.	Strict IT–OT segmentation, application allow-listing on engineering workstations, monitoring IEC-104 traffic. [8]
2023	Supply chain (backdoored SW)	Pipeline monitoring / SCADA	Compromised vendor updates inserted backdoor for remote SCADA access.	Signed/verified updates, strict OT change mgmt, zero-trust for servers/SCADA. [9]
2024	Zero-day exploit	Offshore platform control systems	Remote exploit of zero-day vuln for high-priv access & process control influence.	Harden/allow-list hosts, segment OT networks, OT anomaly detection. [10]
2025	Insider threat	Refinery control systems	Insider modified ICS settings, causing near-miss safety event.	Enforce least privilege and RBAC, apply UEBA monitoring, implement insider-risk programs. [11]

Table 1: Historical Cyber Attacks on ICS

6 Industrial Visit: Roots Cast Private Limited

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Date: December 23, 2025

The team conducted an industrial visit to **Roots Cast Private Limited** to analyze the practical implementation of Industrial Control Systems in a large-scale manufacturing environment. The visit provided critical insights into the "Air-Gapped" security models currently used by legacy industries.

- **Operational Integrity:** Observed how network isolation prevents external intrusions but leaves the system vulnerable to insider logic tampering.
- **Legacy Constraints:** Witnessed the low-compute power of existing PLC hardware, reinforcing the need for our proposed *lightweight* detection framework.
- **Physical-Digital Correlation:** Validated that physical anomalies (vibration/heat) are often checked manually, supporting our proposal for an automated *Physics-Aware* Coordinator model.

7 Proposed Methodology: Hierarchical State-Based Monitoring

Based on the vulnerabilities identified during our simulation, we propose a dual-layer hierarchical monitoring architecture designed to detect anomalies by correlating internal controller logic with the physical laws of the process.

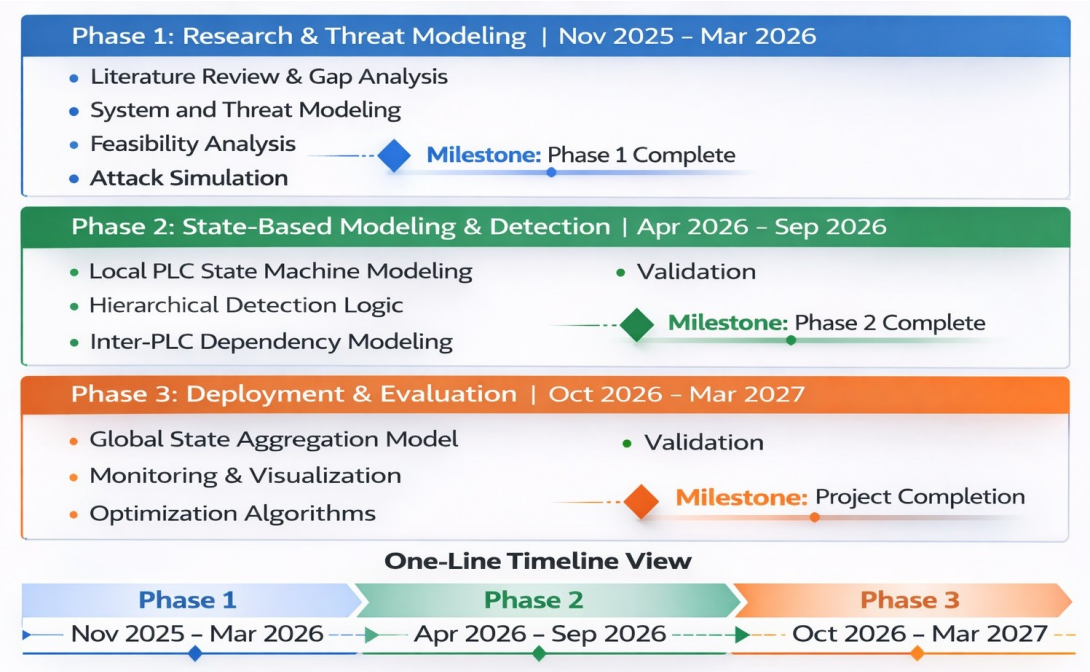
7.1 Layer 1: Local State Model (Controller Level)

This layer resides near the PLC and monitors the **Logic Integrity**. Using a State Transition model, it validates that the internal state transitions follow the programmed sequence. *Detection Case:* If a PLC jumps from an *Idle* state to a *High Pressure* state without passing through the required *Ramping* sequence, an "Illegal Transition" alarm is triggered.

7.2 Layer 2: Global Model (Coordinator)

The Coordinator Model monitors the **Physical Consistency** across multiple assets. It utilizes the relationship between various PLCs (e.g., Input Pump vs. Output Tank) to verify if the reported data obeys the laws of physics. *Detection Case:* If all local PLCs are individually valid, but the relationship between two or more PLCs is violated, the Coordinator Model flags the anomaly and triggers an alarm.

8 Project Timeline



9 Practical Implementation: Protocol Analysis & Attack Simulation

To establish a foundational understanding of Industrial Control System (ICS) vulnerabilities, we performed a practical exploitation analysis using the **Modbus TCP/IP** protocol. This phase was critical in identifying how attackers can manipulate process states without triggering network-layer alarms.

9.1 Modbus Register Interaction via Pymodbus

We developed a Python-based utility utilizing the `pymodbus` library to interact directly with Programmable Logic Controllers (PLCs). The implementation focused on:

- **Digital State Manipulation:** Executing Read/Write operations on *Coils* and *Discrete Inputs* to toggle physical actuators.
- **Analog Data Forgery:** Modifying *Holding Registers* and *Input Registers* to simulate fluctuating sensor data (e.g., pressure or temperature).

9.2 Attack Simulation Outcomes

By simulating Packet Injection attacks, we confirmed that standard PLCs lack inherent authentication for Modbus packets. This allows an adversary to decouple the digital state from the physical process, creating a ”Semantic Gap” that traditional firewalls fail to detect.

10 Current Status & Plan

10.1 Key Deliverables

- Baseline model architecture and implementation.
- Prototype on embedded PLC testbed; validation under attack scenarios.

10.2 Expected Impact

- Framework ready for adoption in Oil & Gas, water utilities, and power grids.
- Addresses critical research gap in real-time threat detection for legacy OT systems.
- Bridges advanced cybersecurity with practical ICS constraints.

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